

Layer Collection

The LayerCollection is a Scriptable object holding the necessary layers for the Advanced Building Systems scripts.



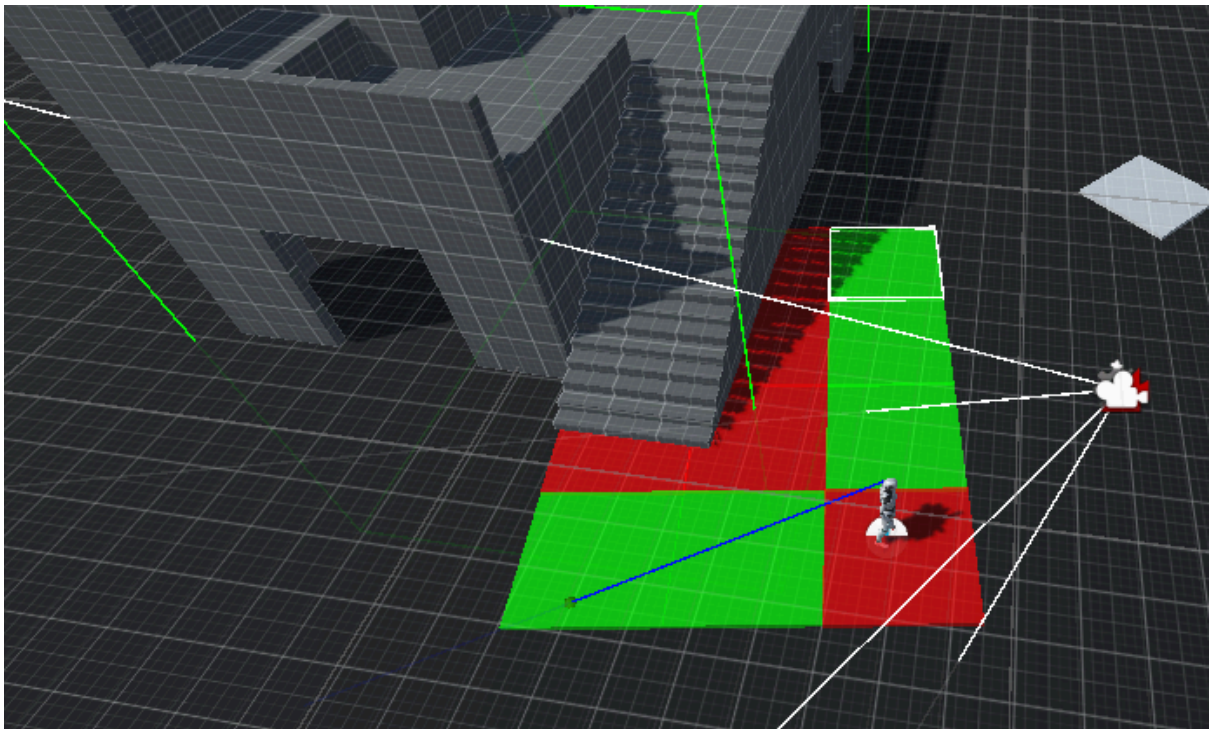
- Layer Of Building Element
 - Only 1 layer can be set to ensure that the system works properly!
 - This layer will be used for the BuildingElements. Make sure your BuildingElements are on this layer.
- Layer Of Player
 - Only 1 layer can be set to ensure that the system works properly!
 - It is used to identify the Player.
 - It is used for blocking the elements during the build process if they are colliding with the player.
 - Also it is used for the [PreBuiltBuildingElement](#) feature.
- Layer Of Ground
 - It is used to identify the ground.
 - For example during the align to ground kind of features and underground check validation.
- Blocking Layer
 - Work similar to the Layer Of the Player but it is only used for blocking the elements if they are colliding with them during the build process.
- Raycast hit Layers
 - It is used for the raycast. What can be hit by the raycast.
 - Usually it is the mix of the Ground and the BuildingElement Layers.

Building Area

The goal of the Building Area Feature is providing the possibility to the developers to create areas in the scene where they can manage the building of the elements. So basically it gives more possibility to the developers to create specific scenarios and manage the behavior of the Advanced Building System like:

- An element can be built near to something, for example an electrical element can be only built near to an electric source.
- Some elements can not be built near to something, for example a wooden element can not be built near the fire.

The main functionality of the BuildingAreas to allow or deny the building of the element.

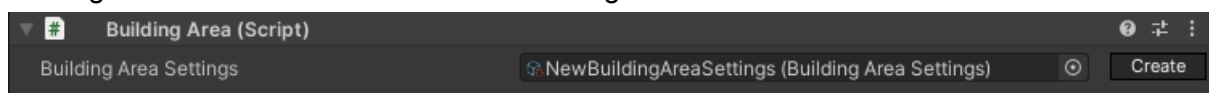


The Building Area created by the following components:

- Building Area (Script component)
- Building Area Settings (Scriptable Object)
- Building Area Ruleset (Scriptable Object)

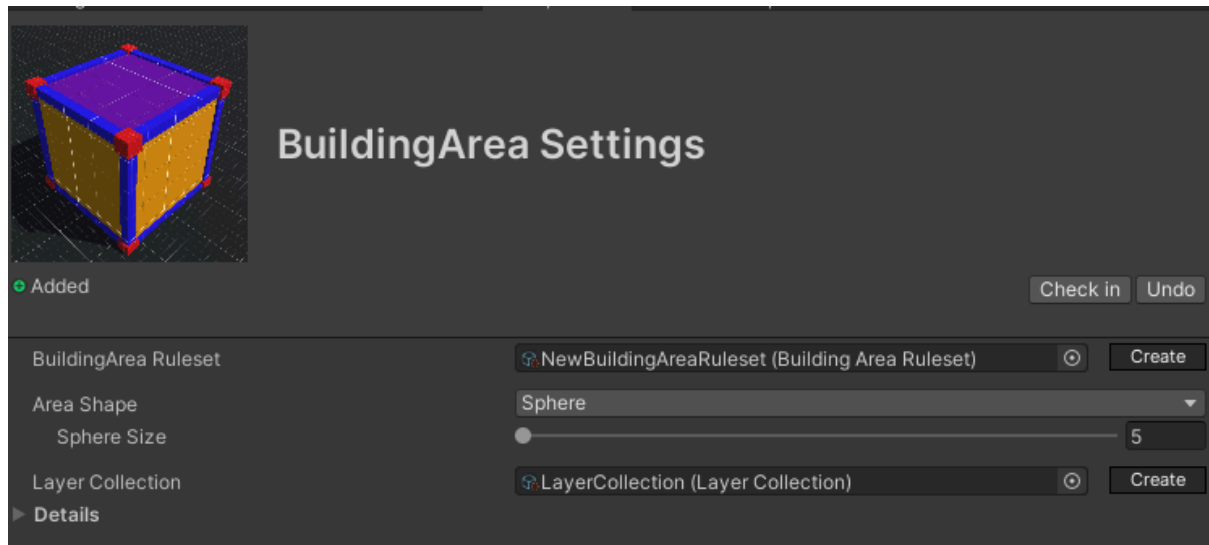
Building Area Component

The script component that is held by the GameObject's should have special area features. In the Awake function the Area will create a trigger collider and allow or deny the element's building inside that collider based on the settings.



Building Area Settings

The goal of the settings scriptable object is to give the possibility to reuse the same area in different situations so the developer doesn't have to set the same properties everytime.



BuildingArea Ruleset : The rules of the area. The allow and deny functionality based on that ruleset.

Area Shape : The shape of the area's collider. Currently 2 shapes are supported.

- Box
- Sphere

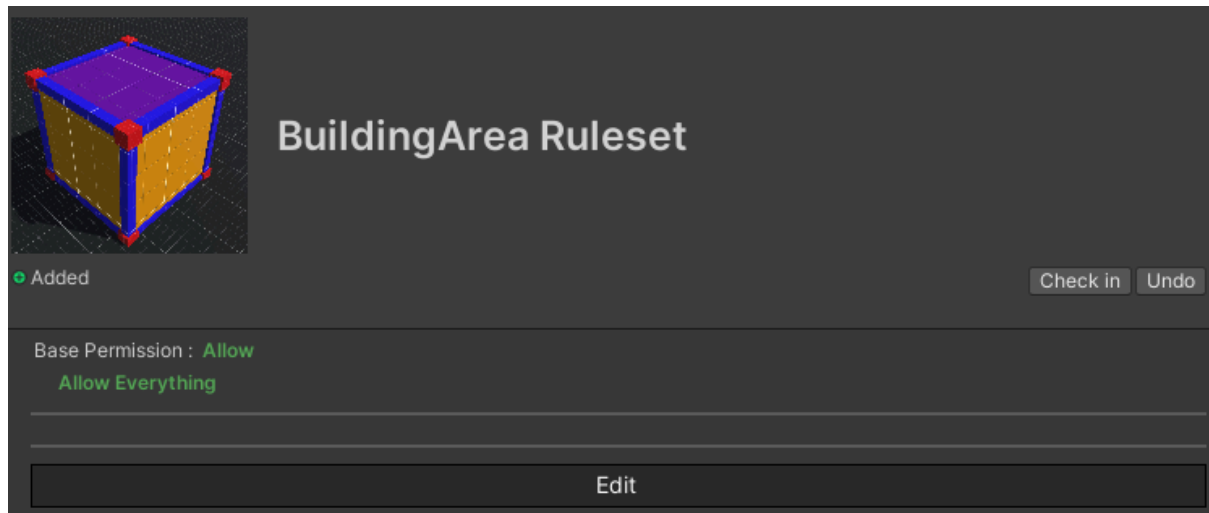
Sphere Size (If the shape is sphere) : The size/radius of the sphere collider created by the Building Area.

Box Size (If the shape is box) : The size of the box collider created by the Building Area in Vector3.

Layer Collection : The layer collection used by the Area.

Building Area Ruleset

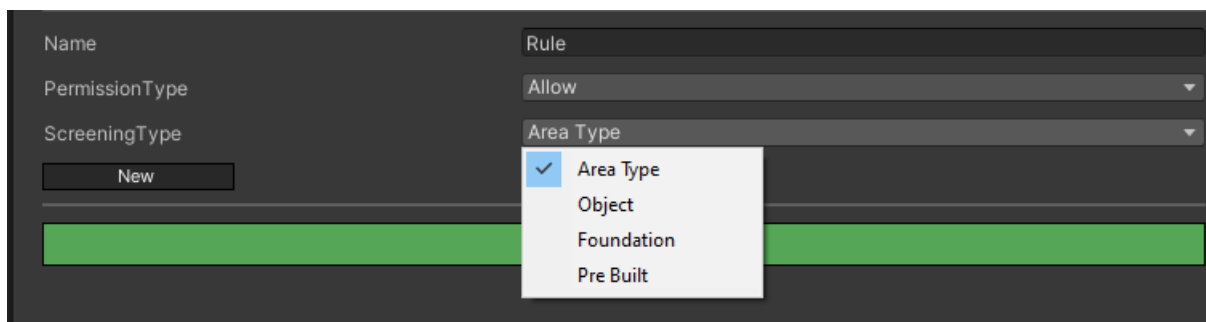
The goal of the ruleset scriptable object is to give the possibility to reuse the ruleset and give the possibility to use the same ruleset with different shaped areas on different areas. The goal is to give the maximum reusability and customizability to the Area feature of the Advanced Building System.



Base Permission: The base permission which will apply on every element what don't included by any of the rules. For example if the Base Permission is Deny and none of the rules said anything about the element (not allowed or denied that element) the building of that element will be blocked inside the area.

BuildingArea Rules have a PermissionType and a ScreeningType.

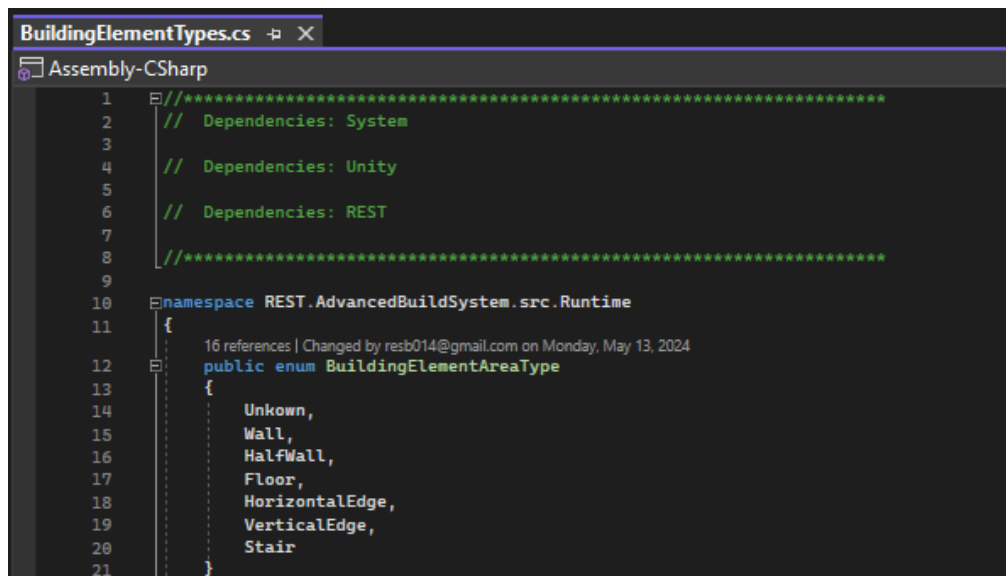
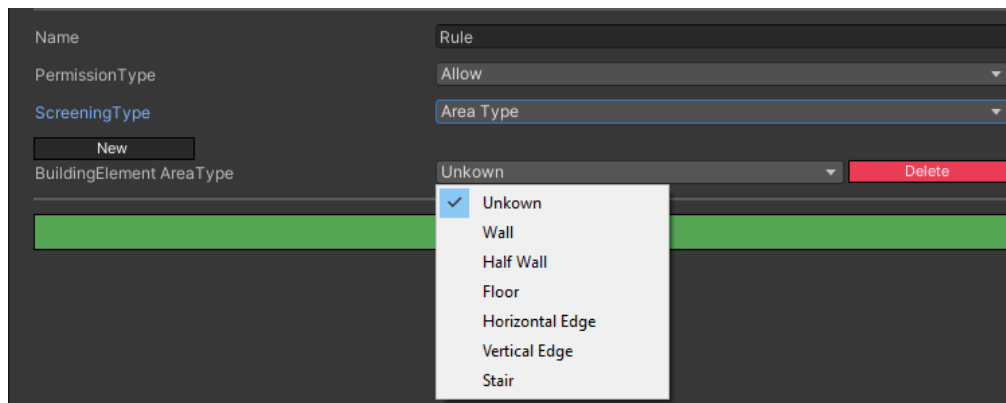
- Name
 - The main purposes are to identify the rule itself and make it easier to work with by the developer.
- Permission Type
 - Allow or Deny the elements that fit into the requirements of the rule.
- Screening Type
 - What should the Area check on the element to decide that it should allow or deny based on the Permission Type.



Area Type

The BuildingElement Area Type is an enum value that is defined in the BuildingElementTypes.cs. You the developer can change these enum values to create your logic. This is meant to identify the elements based on a fully independent type.

Note that every time when the AdvancedBuildingSystem asset is reimported this document can be overridden so ensure that you have a backup from your changes or just do not import this file again.



Building Element (Script)

BuildingElement MetaData: URP_AdvancedGridDemo_Wall_2.5 (Building) [Create]

Prefab Guid: 6dbc12f9-9e50-43a6-9f94-1b6fa08f [New Prefab Guid]

Fixed InstanceGuid: ☐

Final Element: None (Building Element) [v]

Snap To PreBuilt Final Element: ☐

Pre Built: ☒

Foundation: ☐

Area Type: Unkown [v]

Should Allowed By Area: ☐

Should Override: ☐

Indestructible: ☐

Highlight Properties

Highlight Strategy: Automatic Collection [v]

AdvancedGrid

URP_AdvancedGridDemo_Wall_2.5(Clone)

Snap to Grid (Advanced)

Object

If the Rule is set as an Object a set of BuildingElements can be defined. The defined elements will be denied or allowed by the Area.

Name: Rule

PermissionType: Allow [v]

ScreeningType: Object [v]

[New]

BuildingElement Object	URP_AdvancedGridDemo_Corner_2.5 (Building E [v])	Delete
BuildingElement Object	URP_AdvancedGridDemo_Floor_2.5 (Building Ele [v])	Delete

[Save]

Foundation

It simply denies or allows the building of the elements that are signed as Foundation.

Name	Rule
PermissionType	Allow
ScreeningType	Foundation
Save	

PreBuilt

Just like the Foundation logic denies or allows the building of the elements that are signed as [PreBuilt](#).

Name	Rule
PermissionType	Allow
ScreeningType	Pre Built
Save	

The Relationship of the Rules.

In the Editor view you can move the rules Up and Down or Top and Bottom. It is important because the order of the checks are from top to bottom.

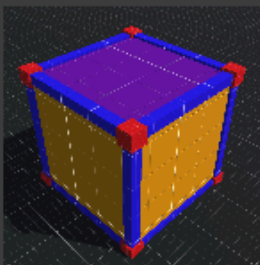
The area's behavior is that:

- If the rule is related to the element and allows it the element will be allowed.
- If the rule is related to the element and Denies it the element will be denied.
- If the rule is not related to the element then the next rule will be checked.

So if the rule is related then the element will be allowed or denied immediately when the rule says any of the two options. For example if the first rule allows the element and the second one denies it the element will be allowed.

Again, the order of the check is from top to bottom.

If all of the rules are unrelated to the element then the base permission will be applied.



BuildingArea Ruleset

Checkout

Base Permission : **Deny** Allow
Deny Everything

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : AreaType

(1) Wall

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Deny**

CollectionType : Object

0 :  URP_AdvancedGridDemo_Corner_2.5 | cc116ff3-a3a1-47d8-9014-4abbf0beed56

1 :  URP_AdvancedGridDemo_Floor_2.5 | 91ac1293-08c9-4c58-b080-19a8f9108bce

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : Foundation

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : PreBuilt

New Entity

Save